

Regime-Closure Epistemology (RCE): A General Principle of Cross-Domain Validation in Coupled Multi-Scale Systems

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Abstract. We propose Regime-Closure Epistemology (RCE) as a general operational principle of validation for coupled multi-scale systems. The principle states that any validation procedure applied to a system with $N \geq 2$ coupled subsystems operating at scale-separated timescales is valid if and only if five joint conditions hold: regime locality, variety closure (Ashby), modelling closure (Conant–Ashby), frozen-parameter governance, and external arbitration via sealed predictions (Beer). When these conditions fail, the system response is the postulation of mystery entities (the “dark X” symptom) whose role is to close ledgers the validation procedure cannot close mechanically. We argue that the Energy-Flow Cosmology (EFC) programme constitutes the first complete operational instantiation of the principle in the empirical sciences, and we develop pre-registered instantiations in medicine (M_0 – M_3 regime architecture), economics (E_0 – E_3), and metrology. The principle is convergent with the cybernetic tradition (Ashby 1956, Conant–Ashby 1970, Bateson 1972, Beer 1972), with second-order cybernetics (von Foerster 1974), and with Friston’s free-energy principle (2010), but adds the operational machinery these traditions lacked: sealed numerical predictions, deposit hashes, and a living falsification ledger. We label the result a *principle*, not a *theorem*, because it characterises operational conditions for validity rather than deriving them from axioms by formal proof. Five falsification conditions are stated for the principle itself; seven pre-registered cross-domain predictions with explicit thresholds and arbiters are deposited against this paper’s DOI. The cross-domain claim is restricted to *homologous validation pathologies*, not ontological identity across domains. The author makes no claim that RCE is correct, only that it is sufficiently precise to be tested, and declares the tests by which it will be.

Keywords: regime-closure epistemology; RCMP; EBE; multi-scale validation; cross-domain isomorphism; cybernetics; frozen-parameter governance; Energy-Flow Cosmology

1 Introduction: the universal dark-matter symptom

A single structural pathology recurs across every empirical domain that attempts to validate models of coupled multi-scale systems. The pathology has different names in different fields. In cosmology it is called *dark matter* and *dark energy*. In medicine it is called *idiopathic*, *essential*, or *functional* disease. In economics it is called *exogenous shocks*, *irrational actors*, or unexplained *risk premia*. In metrology it is called *systematic error* of unknown origin. In climate science it is *internal variability*. In neuroscience and psychiatry it is the categorical residue of the DSM. In AI alignment it is *emergent capability* and *deceptive alignment*.

The pathology is always the same. A field postulates a mystery entity or a residual category to close a quantitative or causal ledger that the field’s own single-regime methodology cannot close. The mystery entity is not predicted by the field’s mechanism; it is required by the field’s measurement architecture. When measurement architecture respects the regime structure of the underlying system, the mystery entity dissolves and is replaced by a coupling, a buffering loop, or a scale-transition operator that was structurally inferable from the beginning.

This paper formalises that observation as a general principle. The principle is not new in spirit — it is convergent with the mid-twentieth-century cybernetic tradition of Ashby, Conant–Ashby, Bateson, and Beer, with second-order cybernetics of von Foerster, and with Friston’s free-energy principle — but it has not previously been stated in a form that admits sealed pre-registered falsification across domains. The Energy-Flow Cosmology (EFC) programme has, over the period 2024–2026, accumulated the structural machinery necessary to make the statement precise. That machinery comprises Entropy-Bounded Empiricism (EBE), the Regime-Consistent Measurement Principle (RCMP), the L0–L3 regime architecture, frozen-parameter governance, sealed numerical predictions with deposit hashes, and a living validation ledger with documented falsifications. We collect these into a single epistemic principle and demonstrate its instantiation in domains other than cosmology.

The author’s working hypothesis, which this paper makes formally explicit, is that the structural problems of cosmology, medicine, economics, metrology, climate science, and AI alignment exhibit *homologous validation pathologies* arising from a common structural cause, rather than being identical problems. The shared cause is the application of single-regime inference to coupled multi-regime systems with nested feedback and buffering. Homology is structural and operational, not ontological — the mechanisms differ across domains, but the failure pattern and its diagnostic signature are the same. The mystery entities are not features of the systems being studied; they are artefacts of the measurement architectures being deployed.

1.1 Position with respect to the EFC corpus

This paper extends and unifies prior work in the author’s corpus:

- Regime-Consistent Measurement Principle (RCMP), DOI 10.6084/m9.figshare.31222900
- Entropy-Bounded Empiricism (EBE) Core Principles, DOI 10.6084/m9.figshare.31222903
- L0–L3 Regime Architecture, DOI 10.6084/m9.figshare.31112536
- Foundations of Energy-Flow Cosmology, DOI 10.6084/m9.figshare.31135597
- Regime-Bound Measurement in Complex Systems, DOI 10.6084/m9.figshare.31564123
- Regime-Locked Measurement as a General Structural Constraint, DOI 10.6084/m9.figshare.31833076
- Regime-Based World Modeling, DOI 10.6084/m9.figshare.31223650
- Validity-Aware AI, DOI 10.6084/m9.figshare.31122970
- Cross-Domain Bridge Equations, DOI 10.6084/m9.figshare.31940547
- KT3b Cross-Regime Measurement Failure, DOI 10.6084/m9.figshare.31963821
- Composite-Level Aggregation paper, DOI 10.6084/m9.figshare.32162706
- Methodological Convergence in Stage-IV Cosmology, DOI 10.6084/m9.figshare.32080080

These prior papers each address one facet of the structure. This paper states the underlying principle, names it Regime-Closure Epistemology (RCE), and demonstrates that EFC is one instantiation among several the principle admits.

1.2 Scope discipline

This paper makes a strong claim — that a single epistemic principle governs validation across most of empirical science. Such claims are properly suspect. The discipline imposed here, mirroring EFC practice, is fourfold: (i) the principle is stated in a form that admits falsification; (ii) each cross-domain instantiation is given pre-registered, sealed predictions with explicit failure thresholds; (iii) the structural isomorphism between domains is made mathematically explicit, not metaphorical; and (iv) the author’s prior falsifications within EFC (EFC-BIC v0.1 bar-instability prediction, original ISW cancellation metric, naive GRAV-to-cosmology mapping, monotonic

radial boost in EFC v1) are preserved in the validation ledger and serve as evidence that the apparatus does in fact reject claims it cannot support.

The paper is not a claim that the principle is correct. It is a claim that the principle is sufficiently formalised to be tested, and a declaration of the tests by which it will be tested.

2 Inherited foundations

This section recapitulates the structural machinery inherited from the EFC corpus. Readers familiar with RCMP, EBE, and the L0–L3 architecture may skip to Section 3.

2.1 Entropy-Bounded Empiricism (EBE)

EBE (DOI 10.6084/m9.figshare.31222903) is the methodological constitution of EFC. Its core commitments are:

- **Regime priority.** Every observable, every parameter, and every prediction is tagged with the regime in which it is physically active. Cross-regime inference requires an explicit coupling operator, not an implicit assumption of universality.
- **Falsifiability over plasticity.** Every prediction is paired with an explicit falsifier. Theory extensions that absorb failures via additional parameters are treated as model extensions, not as validation.
- **Historical preservation.** Falsified predictions are preserved in the ledger with their failure mode documented. Survivorship bias is structurally prohibited.
- **Frozen parameters across regimes.** Parameters that appear in multiple regimes are locked across all of them. Re-tuning per dataset is treated as evidence of model misspecification, not of legitimate fit.

EBE is theory-agnostic. It is a discipline of validation, not a particular physical hypothesis. It is one of the components RCE generalises.

2.2 The Regime-Consistent Measurement Principle (RCMP)

RCMP (DOI 10.6084/m9.figshare.31222900) states that an observable can only constrain mechanisms active in the regime where the observable is generated. The principle has both a positive and a negative form.

Positive form: predictions made about regime R_i are validated by data drawn from R_i under measurement conditions appropriate to R_i .

Negative form: data drawn from regime R_i *cannot* validate or falsify predictions about regime R_j ($j \neq i$) without an explicit, derived coupling operator linking the two regimes. Apparent inconsistencies under cross-regime inference are systematically ambiguous — they may reflect failure of the model, failure of the coupling, or failure of the measurement, and the three cannot be disentangled without independent constraints in each regime.

The KT3b note (DOI 10.6084/m9.figshare.31963821) is a worked example of RCMP-failure: a quantity (environmental density gradient ρ_{env}) measured at the L1/L2 boundary cannot constrain a mechanism D_{eff} defined in L2 without an explicit $\alpha(\rho_{\text{env}})$ operator; without the operator, the null result is uninterpretable.

2.3 L0–L3 regime architecture

The L0–L3 architecture (DOI 10.6084/m9.figshare.31112536) is the cosmological instantiation of regime typology. In its general form, an L_0 – L_n architecture for any complex system is constructed from three ingredients:

1. **Characteristic scales.** A set of physical, biological, or institutional scales $\{\lambda_i\}$ separated by ratios $\lambda_{i+1}/\lambda_i \geq K$ for some scale-separation constant $K > 1$. The separation can be spatial, temporal, energetic, or organisational.

2. **Characteristic dynamics.** Each scale λ_i admits a distinct effective theory $\mathcal{T}_i$ with its own degrees of freedom, its own conservation laws, and its own equations of motion. The effective theories are not reducible to one another except via explicit coarse-graining or refinement operators.
3. **Coupling operators.** Adjacent regimes are linked by operators $\mathcal{C}_{i,i+1}$ that map the state space of $\mathcal{T}_i$ into the state space of $\mathcal{T}_{i+1}$. These operators are typically not identity maps; they are where buffering, emergence, and screening occur.

In cosmology, the regimes are L_0 (early universe, linear), L_1 (transition), L_2 (structure formation, weak nonlinearity), L_3 (galactic and sub-galactic, fully nonlinear). The architecture generalises trivially to any domain admitting clear scale separation (Sections 4.2–4.5).

2.4 Frozen-parameter governance

A parameter θ that appears in multiple regimes is governed by the constraint

$$\theta|_{R_i} = \theta|_{R_j} \quad \forall i, j.$$

This is the formal statement of frozen-parameter governance. Its function is to prevent the most common failure mode in multi-regime modelling: the silent reabsorption of failed predictions via per-regime tuning. Frozen-parameter governance is the empirical-side analogue of the Conant–Ashby modelling closure theorem (Section 2.5).

2.5 The cybernetic substrate

The author was unaware, at the time of formulating EBE and RCMP, that substantially equivalent statements existed in the cybernetic tradition. The convergence is, however, evidence *for* the underlying structure rather than against the originality of EFC’s formulation: independent derivation from distinct empirical contexts strengthens the case that the structure tracks something real about multi-regime systems.

The relevant prior results are:

- **Ashby’s Law of Requisite Variety (1956).** A regulator R of a system S subject to disturbance D can stabilise S only if the variety (state-space size) of R satisfies $V(R) \geq V(D)/V(B)$ where $V(B)$ is buffering variety. Applied to validation: a measurement architecture can validate a multi-regime system only if the architecture’s own variety matches the system’s.
- **Conant–Ashby Theorem (1970).** Every good regulator of a system must contain a model of that system. Formally, if $H(Z | R, E)$ denotes the conditional entropy of system outcome Z given regulator output R and environment E , then optimal regulation requires the regulator to embody a structural model $M(E)$ of the environment such that $H(Z | R, M(E))$ is minimised over admissible R . Applied to validation: a validation pipeline must contain a structural model of *each regime* of the system being validated, not merely of the system as a single object.
- **Bateson’s logical types (1972).** Confusion of logical types — treating a statement about the system as if it were a statement within the system — is the structural origin of a broad class of pathologies. Applied here: confusion of regime level is the structural origin of mystery-entity postulation.
- **Beer’s Viable System Model (1972).** A complex system capable of homeostatic survival requires explicit external arbitration channels (algedonic channels) that bypass internal narrative coherence. Applied to validation: sealed, pre-registered predictions function as algedonic channels for scientific theories, preventing internal narrative absorption of failed predictions.

- **Friston’s free-energy principle (2010).** Any self-organising system maintains itself by minimising a variational free energy $F = E_q[\ln q(s) - \ln p(o, s)]$ over states s and observations o . The principle is, formally, second-order cybernetics expressed in Bayesian language. The Cross-Domain Bridge Equations paper (DOI 10.6084/m9.figshare.31940547) establishes that EFC’s gradient-flow dynamics $dF/dt \leq 0$ is structurally identical.

The cybernetic tradition supplied the theorems. EFC supplies the first instantiation with sealed numerical predictions, deposit hashes, and a falsification ledger. The synthesis admits a stronger statement than either tradition alone has made.

3 The General Principle: Regime-Closure Epistemology

3.1 Status: principle, not theorem

A terminological note before the statement. We deliberately label the following result a *principle*, not a *theorem*. The distinction is not cosmetic. A principle in the strict mathematical sense is derived from axioms by formal proof. The following statement is not of that kind: it is a characterisation of necessary and sufficient operational conditions for validation procedures in coupled multi-regime systems, where “valid” is defined relative to an empirical standard (production of replicable, falsifiable predictions whose failure modes are documented) rather than from first principles. This places the result in the same epistemic category as RCMP (Regime-Consistent Measurement *Principle*) and EBE (Entropy-Bounded *Empiricism*), not in the category of, say, Conant–Ashby’s modelling theorem or Ashby’s variety inequality, both of which are formally proven. The principle takes those theorems as load-bearing components but is itself an operational principle.

3.2 Statement

Principle (Regime-Closure Epistemology, RCE). Let S be a complex system composed of $N \geq 2$ coupled subsystems $\{S_i\}_{i=1}^N$, where each S_i operates at characteristic timescale τ_i or characteristic scale λ_i , satisfying a scale-separation condition $\tau_{i+1}/\tau_i \geq K$ for some constant $K > 1$. Let $\{S_i\}$ be coupled by inter-regime operators $\{\mathcal{C}_{i,j}\}$. Let V be a validation procedure applied to a model M of S . Then V is a *valid* procedure for M if and only if the following five conditions are jointly satisfied:

- Regime locality.** Each prediction π_i of M pertaining to subsystem S_i is tested against observables $\mathcal{O}_i$ generated within S_i under measurement conditions appropriate to S_i . Cross-regime inference requires explicit coupling operators $\mathcal{C}_{i,j}$ that are themselves derived or independently constrained.
- Variety closure (Ashby).** The validation pipeline V has variety

$$V(V) \geq V(S) / V(B),$$

where $V(S)$ is the state-space variety of S and $V(B)$ is the buffering variety of S . Below this threshold, V systematically underconstrains M .

- Modelling closure (Conant–Ashby).** V contains a structural model of every S_i and of every active coupling operator $\mathcal{C}_{i,j}$. Validation absent such a model produces guaranteed systematic bias proportional to the coupling strength.
- Frozen-parameter governance.** Every parameter θ that appears in two or more regimes is locked, $\theta|_{R_i} = \theta|_{R_j} \forall i, j$. Per-regime re-tuning is mathematically equivalent to model extension and must be declared as such.

- (e) **External arbitration (Beer).** V employs at least one channel of sealed, pre-registered prediction whose outcome is determined externally to the validation pipeline. Internal-only validation is structurally insufficient because sufficiently complex systems can narratively absorb arbitrarily discrepant results.

When (a)–(e) jointly fail, the symptom is the postulation of mystery entities (the “dark X” phenomenon) whose role is to close the ledger that the validation procedure has been forbidden from closing mechanically.

3.3 Corollary: the mystery-entity diagnosis

Corollary 1 (Universal symptom). Let $\mathcal{X}$ be a domain that has produced a stable mystery entity (dark matter, idiopathic disease, exogenous shock, internal variability, etc.) used to close a quantitative or causal ledger. Then $\mathcal{X}$ is, with high prior probability, validating a multi-regime system under a single-regime measurement architecture that violates one or more of conditions (a)–(e).

Corollary 2 (Dissolution). If the validation architecture of $\mathcal{X}$ is upgraded to satisfy (a)–(e), then either (i) the mystery entity becomes empirically testable and either survives as a real entity or is falsified, or (ii) the mystery entity is replaced by an explicit coupling operator $\mathcal{C}_{i,j}$ or buffering mechanism, with the residual variance absorbed by the operator.

3.4 Why the failure produces “dark” postulates specifically

Condition (b), Ashby variety closure, makes the dark-postulate symptom mathematically inevitable. When $V(V) < V(S)/V(B)$, the validation pipeline systematically under-resolves the system. The under-resolved component cannot be measured but must be accounted for in any closed ledger (energy, mass, growth, prevalence, expected return). The under-resolved component is therefore named, given symbolic mass, and treated as a hidden entity. Its function is mathematical bookkeeping, not empirical reference.

This explains why “dark X” postulates share certain features across domains: they have approximately the right magnitude to close the ledger; they are by construction not directly observable; and their properties are determined by what the ledger requires them to be rather than by independent measurement. These are precisely the features of under-resolution residuals, not the features of real entities.

3.5 Distinguishing residual categories: a necessary caveat

The “dark X” framing is operationally useful but must not be allowed to collapse three distinct epistemic categories into one. We distinguish:

- **Empirically characterised, ontologically unidentified.** Dark matter as currently constituted: its gravitational signature is precisely measured (rotation curves, lensing, CMB acoustic peaks, structure formation), but its particle content remains unknown after decades of direct-detection campaigns. The empirical characterisation is real; the ontology is not.
- **Descriptive residual categories.** Medical “idiopathic” or “essential” labels: a category whose content is exactly its failure of mechanism-attribution. The empirical characterisation is weaker; the category is closer to a placeholder than to a measured signature.
- **Explicit modelling residuals.** Knightian uncertainty in economics; “internal variability” in climate ensembles: residuals that are honestly declared as outside the model’s scope, with no claim that they correspond to a hidden entity.

The RCE diagnosis claims these three categories share a *structural homology*: each arises when single-regime inference is applied to a multi-regime system. They are not claimed to be ontologically identical. A precisely characterised gravitational signature is not the same epistemic

object as a clinical placeholder label. What is the same is the structural cause of the gap they fill in the field’s ledger.

This distinction matters for the dissolution prediction in Corollary 2. RCE-compliant validation in cosmology may reveal that dark matter is partially a real entity (a particle yet to be detected) *and* partially a regime-closure artefact (the fraction explained by entropy-gradient gravitational response in EFC). RCE-compliant validation in medicine is more likely to reveal that idiopathic categories are predominantly regime-closure artefacts, because the descriptive-residual category has weaker empirical anchoring to begin with. The principle does not predict that all mystery entities are artefacts. It predicts that the fraction explained by regime closure is presently underestimated across these domains.

4 Domain instantiations

This section instantiates the principle in five domains. Cosmology is treated as the proof-of-concept case (the case where the EFC programme has demonstrated all five conditions can be operationally enforced). Medicine, economics, metrology, and AI alignment are treated as candidate domains where the principle predicts (i) the location of the mystery-entity symptom and (ii) the structure of the regime-closure upgrade.

4.1 Cosmology (proof-of-concept)

EFC is the case for which the principle was first developed. Its regime architecture is L_0 – L_3 as defined in Section 2.3. The “dark X” postulates of standard cosmology are dark matter and dark energy. Under the RCE diagnosis, these are under-resolution residuals of single-regime inference: the standard cosmological pipeline (Λ CDM) applies the same parametric scheme across the early universe, structure formation, and galactic regimes, with no explicit regime-transition operators.

The EFC instantiation enforces all five conditions:

- Condition (a): regime tagging on every observable; cross-regime leakage prohibited (RCMP, DOI 31222900).
- Condition (b): the validation pipeline maintains state for background, perturbation, lensing, and discrete-gravity sectors as independent regulators; the Framework Atlas tracks 43 frameworks $\times$ 35 phenomena $\times$ 197 predictions, which is the variety-matching infrastructure.
- Condition (c): every parameter is mapped to its operational regime in the Global Parameter Registry, and inter-regime coupling operators ($T(a)$, $\Theta(\rho)$, $\Sigma_{\text{eff}}(z)$) are explicit.
- Condition (d): frozen-parameter governance enforced; falsifications of attempted re-tuning preserved in the ledger (e.g. Variant H MCMC $\Delta\text{AIC} = +4.05$).
- Condition (e): pre-registered sealed predictions P1–P8 with deposit hashes and Stage-IV arbiters (Euclid DR1, Rubin DP2, DESI DR2/DR3, SO $\times$ Euclid).

The structural reorganisation of EFC over 2024–2026 — from a unified-gravity hypothesis to a regime-closure architecture in which the background sector is mostly empty and the perturbation sector carries the signal — is an example of the principle applied to itself: the mystery entities (*dark* matter, *dark* energy) are being progressively replaced by explicit coupling operators ($\mu(a)$, $\Sigma(z)$, $\eta(k, z)$) within a regime-locked validation architecture.

4.2 Medicine: the M0–M3 architecture

Scope and stringency caveat. The medical instantiation in this subsection is a *structural skeleton*, not an empirical validation of EFC-equivalent stringency. The author holds no medical credentials. The regime architecture proposed below is intended as a draft to be corrected by domain experts, and the pre-registered predictions in Section 7.1 are structural inferences from the principle rather than clinical hypotheses derived from independent biomedical evidence. The

operational parameters, derived coupling operators, and sealed predictions that would constitute an EFC-grade medical instantiation require a dedicated follow-up paper. What is deposited here is the architectural skeleton, the pre-registered prediction structure, and the falsification criteria; the empirical fleshing-out is left to subsequent work.

Mystery entities. Medicine generates an unusually clean set of dark-postulates: *idiopathic* (cause unknown), *essential* (cause unknown, restated), *functional* (cause unknown, located in function rather than structure), *psychosomatic* (cause unknown, located in psyche), *autoimmune of unknown trigger*, *central sensitisation*, *neuroinflammation of unknown origin*. The prevalence and burden of conditions in these categories (long COVID, ME/CFS, fibromyalgia, IBS, atopic multi-morbidity, treatment-resistant depression, idiopathic autoimmunity, chronic inflammatory states) is substantial and growing.

Diagnosis under RCE. Medicine systematically violates conditions (a)–(c) of the principle by organisational design. Specialist silos — cardiology, endocrinology, immunology, gastroenterology, psychiatry, neurology — map directly onto physiological subsystems that are nevertheless coupled by multiple feedback loops on overlapping timescales. A patient with dysregulation distributed across HPA-axis, autonomic balance, gut microbiota, immune tone, and circadian rhythm presents to one silo at a time; each silo measures within its own regime and either finds nothing actionable or treats locally. The systemic coupling is unmeasured.

This is RCMP violation at clinical scale.

The M0–M3 regime architecture. We propose the following formal regime structure for medicine, by direct analogy with L_0 – L_3 in cosmology:

Regime	Timescale	Active observables
M_0 molecular / genomic	seconds–minutes	enzyme kinetics, signalling cascades, gene expression
M_1 cellular / tissue	minutes–hours	inflammatory markers, hormone pulses, autocrine signalling
M_2 organ / system	hours–days	HRV, cortisol rhythm, insulin sensitivity, autonomic balance
M_3 organism / behaviour	days–years	sleep, social contact, nutrition, identity, habit

Inter-regime coupling operators include the HPA axis ($M_2 \leftrightarrow M_3$), gut–brain axis ($M_1 \leftrightarrow M_3$), allostatic load (McEwen 1998, formally $M_2 \leftrightarrow M_3$ buffer exhaustion), and chronobiological coupling ($M_0 \leftrightarrow M_3$).

Frozen-parameter governance in medicine. A patient’s allostatic state, autonomic tone, and circadian phase are parameters that should be locked across M_0 – M_3 during diagnosis. At present they are typically measured by no specialist and tracked by none. A regime-closure-compliant clinical pipeline would require these to be characterised once and held fixed across all subsequent specialist measurements within the same diagnostic episode.

Mystery-entity dissolution. Under the RCE diagnosis, the prediction is that conditions currently classified as idiopathic are mostly cross-regime dysregulations whose mechanism becomes visible only when measurement architecture respects regime structure. The proposed regime-closure upgrade is testable via the pre-registered predictions in Section 7.1.

4.3 Economics: the E0–E3 architecture

Mystery entities. Economics generates dark-postulates of its own: *exogenous shocks* (cause located outside the model boundary by fiat), *irrational actors* (residual variance attributed to agents), *animal spirits*, *liquidity preference*, and unexplained *risk premia*.

Diagnosis under RCE. Standard macroeconomic modelling (DSGE and its descendants) typically assumes separability between micro-foundations (household and firm decisions), credit

and banking dynamics, and macro-aggregate stabilisation. This is an explicit single-regime architecture imposed on a system with strong nested feedback on disparate timescales. Goodhart’s Law (“when a measure becomes a target, it ceases to be a good measure”) is precisely an RCMP violation: a measurement optimised within one regime drives the system into another regime where the original measurement is no longer valid.

The E0–E3 regime architecture. The economic regime structure parallels the medical and cosmological cases, with characteristic timescales ranging from settlement-level to civilisational:

Regime	Timescale	Active observables
E_0 transactional	seconds–days	order flow, payment settlement, daily liquidity
E_1 business cycle	months–years	inflation, employment, credit growth
E_2 institutional	years–decades	regulatory regimes, financial architecture
E_3 structural / civilisational	decades–centuries	demographic transition, technological paradigm

Inter-regime coupling operators are well-documented but rarely explicitly modelled: credit buffering ($E_0 \leftrightarrow E_1$), institutional path-dependence ($E_1 \leftrightarrow E_2$), and Minsky-style instability accumulation ($E_2 \leftrightarrow E_3$).

Mystery-entity dissolution. Predicted under RCE: a substantial fraction of variance currently attributed to “exogenous shocks” is internally generated by E_0 – E_1 or E_1 – E_2 coupling that exceeds the buffering capacity of the lower regime. Operationally, this is the Minsky financial-instability hypothesis stated as an RCE corollary.

4.4 Metrology: the instrument-coupling case

Metrology is a structurally distinct instantiation of the principle because the validation procedure V is itself the system S being analysed: the instrument and the measurand are coupled. The Regime-Bound Measurement paper (DOI 10.6084/m9.figshare.31564123) and the Regime-Locked Measurement as General Structural Constraint paper (DOI 10.6084/m9.figshare.31833076) develop this case in detail.

The mystery entity is *systematic error of unknown origin*. Under RCE diagnosis, systematic errors of unknown origin are exactly RCMP violations: the instrument is operating outside its calibrated regime, or the measurand has crossed a regime boundary the instrument was not designed to track, or instrument and measurand are coupled through a buffer that absorbs the signal of interest.

The metrological instantiation of RCE produces concrete operational guidance: every measurement campaign must (i) tag the regime of the instrument, (ii) tag the regime of the measurand, (iii) declare any coupling between them, and (iv) freeze instrument-side parameters across the campaign. These are precisely the operational requirements imposed by RCMP/EBE in cosmology.

4.5 AI alignment: the deployment-distribution case

Mystery entities. *Emergent capability, deceptive alignment, specification gaming, Goodhart on proxy reward, distribution shift.*

Diagnosis under RCE. The training regime and the deployment regime of a machine-learning system are distinct L_i in the RCE sense. The training-time loss is an observable in the training regime; the deployment-time behaviour is an observable in the deployment regime. Validation that measures only the former and predicts the latter is a textbook RCMP violation, formally identical to the KT3b cosmology case (DOI 10.6084/m9.figshare.31963821).

The Validity-Aware AI paper (DOI 10.6084/m9.figshare.31122970) develops the AI instantiation. The Cross-Domain Bridge Equations paper (DOI 10.6084/m9.figshare.31940547) develops the formal mapping between EFC’s gradient-flow dynamics and RLHF-as-thermodynamic-

isomorphism (DOI 10.6084/m9.figshare.31940535). Together these establish the AI instantiation of RCE.

4.6 Climate science: brief

Climate models share the same single-regime aggregation pathology (IPCC ensembles average across regime transitions). The instantiation is structurally identical to the cosmological case and is mentioned here without separate development; the operational regime architecture would be ocean–atmosphere–biosphere–cryosphere with explicit timescale-separated coupling operators.

5 Structural isomorphism: formal mapping

The instantiations in Section 4 are not analogies. They are formally isomorphic instances of the same mathematical structure. This section makes the isomorphism explicit.

5.1 Object-level correspondence

Construct	Cosmology	Medicine	Economics
Regimes	L_0-L_3	M_0-M_3	E_0-E_3
Scale-separation	Hubble, structure, virialisation	ms, hour, day, year	second, month, decade, century
Mystery entity	dark matter, dark energy	idiopathic, essential	exogenous shocks
Buffering	screening, $\Theta(\rho)$	allostasis, HPA axis	credit, fiscal capacity
Coupling operator	$T(a), \mu(a), \Sigma(z)$	gut–brain axis, HPA axis	Minsky, Goodhart
External arbiter	Euclid, Rubin, DESI	RCT, longitudinal cohort	natural experiment, historical episode
Frozen parameter	β, z_t, k	allostatic load, autonomic tone	potential GDP, neutral rate

5.2 Failure-mode correspondence

Failure mode	Cosmology instance	in-	Cross-domain instance
Per-regime re-tuning	Refit per-survey α_{L2}		Per-trial dose adjustment without protocol pre-registration; per-quarter DSGE re-estimation
Single-regime aggregation	Composite over inner+outer (32162706)	ΔAIC	Whole-population RCT outcome obscuring subgroup heterogeneity; aggregate GDP obscuring sectoral collapse
Cross-regime inference without operator	KT3b (31963821)		Inferring cellular mechanism from organism-level symptoms; inferring micro from macro without explicit aggregation
Internal narrative absorption	“Tension” relabelled as “feature”		“Heterogeneity” relabelled as “patient-specific”; “model uncertainty” relabelled as “Knightian”

5.3 Governance correspondence

Governance	Cosmology	Medicine	Economics
Pre-registration	Sealed DOI, SHA-256 hash	ClinicalTrials.gov, OSF	EGAP, pre-analysis plan
Frozen parameters	Global parameter registry	Locked allostatic baseline	Locked structural parameters
External arbiter	Stage-IV survey	Independent RCT replication	Out-of-sample forecast
Falsification ledger	EFC validation ledger	Living registry of failed indications	Failed-model registry (rare in practice)

The third row of the governance table is informative: medicine has partial instantiations of RCE governance (ClinicalTrials.gov, RCT culture); economics has weaker partial instantiations (EGAP, pre-analysis plans); neither has the full set. Cosmology, via EFC, has the most complete instantiation to date.

6 Falsification conditions for RCE itself

A principle this general must itself be falsifiable, on pain of being indistinguishable from a sufficiently flexible narrative. Five conditions, jointly or severally, falsify RCE:

- (F1) **Single-regime success in a multi-regime domain.** If a domain governed by genuinely coupled multi-regime dynamics achieves validated predictive success using strictly single-regime

inference (no explicit coupling operators, no frozen-parameter governance across regimes, no external sealed predictions), and the success is replicated independently, the principle is falsified. Operational threshold: predictive $R^2 > 0.5$ on out-of-sample data spanning two or more regimes, achieved without any regime coupling structure in the model.

- (F2) **Regime-closed validation failing systematically.** If a domain adopts RCE-compliant validation (a)–(e) and systematically produces *worse* predictive performance than the single-regime alternative across multiple instantiations, the principle is falsified. Operational threshold: in three or more independent domain implementations, RCE-compliant validation yields larger out-of-sample prediction error than single-regime baseline at $p < 0.05$.
- (F3) **Mystery entities surviving regime closure.** If, in any domain, RCE-compliant validation is implemented in full and the mystery entity that motivated the implementation (dark matter, idiopathic prevalence, exogenous shocks) survives intact rather than dissolving into explicit operators or becoming independently measurable, the principle is weakened (but not immediately falsified, since the entity may be real). Operational threshold: surviving entities must be independently measurable within five years of regime-closure adoption; if not, the dissolution claim of Corollary 2 is falsified for that domain.
- (F4) **Cross-domain isomorphism failure.** If the cross-domain mapping in Section 5 fails to produce transferable predictions — i.e. a prediction structurally derived in one domain (e.g. medicine) does not survive translation to another domain (e.g. economics) following the isomorphism table — the unification claim is falsified, though individual domain instantiations may survive.
- (F5) **Information-theoretic absence of variety constraint.** If Ashby’s variety-matching condition (b) can be shown to be operationally non-binding — i.e. if validation pipelines with $V(V) \ll V(S)/V(B)$ systematically produce equally good predictive performance — the principle’s cybernetic grounding is falsified.

7 Pre-registered predictions

The following predictions are deposited as sealed pre-registrations against this paper’s DOI. Each prediction has an explicit observable, a quantitative threshold, an external arbiter, and a defined failure criterion. The deposit acts as the algedonic channel in Beer’s sense: internal narrative reabsorption is structurally blocked.

7.1 Medicine

P-MED-1 (M2–M3 coupling visibility). In a cohort of patients with currently-idiopathic chronic conditions (long COVID, ME/CFS, fibromyalgia; $n \geq 500$ per condition), simultaneous frozen measurement of M_2 regime parameters (allostatic load index, autonomic balance via 24-hour HRV, cortisol rhythm slope) across all participants will reveal cluster structure with effect size $d \geq 0.5$ that does not appear in conventional single-specialist diagnostic workflows. Falsifier: clusters do not emerge, or emerge with $d < 0.3$. Arbiter: independent replication in two or more cohorts. Confrontation window: ≤ 5 years from deposit.

P-MED-2 (Cross-regime intervention superiority). Therapeutic protocols designed to address cross-regime coupling explicitly (combined M_2/M_3 intervention with frozen-parameter tracking) will outperform best-current-practice single-specialist intervention on idiopathic chronic conditions with relative effect size $\geq 1.4\times$ on validated symptom scales. Falsifier: relative effect size $< 1.1\times$ or significantly negative. Arbiter: pre-registered RCT, $n \geq 200$. Confrontation window: ≤ 7 years from deposit.

P-MED-3 (Variety-matching diagnostic precision). A diagnostic protocol enforcing variety closure (condition b) — i.e. requiring simultaneous measurement across M_0 , M_1 , M_2 , and M_3 regimes with locked baselines — will reduce the “idiopathic” classification rate by $\geq 40\%$

in the conditions above, relative to current standard-of-care diagnostic workflows. Falsifier: reduction $< 15\%$. Arbiter: side-by-side comparison in specialty clinics. Confrontation window: ≤ 5 years from deposit.

7.2 Economics

P-ECON-1 (Coupling-aware crisis prediction). A macroeconomic forecasting protocol enforcing E_0 – E_1 coupling explicitly (real-time credit buffering metrics held as frozen parameters across business-cycle forecasts) will outperform state-of-the-art DSGE benchmarks on out-of-sample crisis-onset forecasting (banking crisis, sovereign crisis, currency crisis), with relative reduction in false-negative rate $\geq 30\%$ at matched false-positive rate. Falsifier: false-negative reduction $< 10\%$. Arbiter: out-of-sample evaluation on post-deposit crisis episodes. Confrontation window: ≤ 10 years from deposit.

P-ECON-2 (Goodhart-rate measurement). A direct measurement of Goodhart-rate (the rate at which a metric under policy targeting decouples from the underlying construct it was designed to measure) will show systematic dependence on regime distance between the targeting regime and the underlying-construct regime. Specifically: Goodhart-rate $\propto |E_i - E_j|$ where the targeting acts at regime E_i and the construct is defined at regime E_j . Falsifier: no significant correlation with regime distance. Arbiter: historical analysis of pre-registered policy episodes. Confrontation window: ≤ 3 years from deposit.

7.3 Metrology

P-MET-1 (Cross-regime calibration deficit). In any precision measurement campaign where instrument-regime and measurand-regime are distinct (e.g. atomic clock in spatially inhomogeneous gravity, neural recording during state transitions, sensor in turbulent versus laminar flow), the residual systematic error attributable to RCMP-violation (cross-regime measurement without explicit coupling operator) will scale as $\sigma_{\text{sys}} \propto V(S)/V(V)$ where $V(S)/V(V)$ is the variety mismatch as defined in Section 3. Falsifier: σ_{sys} is independent of variety mismatch within measurement precision. Arbiter: meta-analysis of precision-measurement campaigns with documented cross-regime structure. Confrontation window: ≤ 5 years from deposit.

7.4 Cross-domain meta-prediction

P-META-1 (Isomorphism transfer). Predictions derived structurally in one domain (e.g. P-MED-3) will admit translation into a structurally equivalent prediction in another domain (e.g. economics) following the isomorphism table in Section 5.1, and the translated prediction will be empirically supported. Specifically: P-MED-3 (variety closure reduces “idiopathic” classification by $\geq 40\%$) translates to “variety closure reduces unexplained variance in macroeconomic crisis prediction by $\geq 30\%$ ” (which is P-ECON-1 in a different form). If P-MED-3 passes and P-ECON-1 fails (or vice versa) while domain conditions are otherwise similar, the isomorphism claim is weakened. If both fail, RCE itself is weakened. Confrontation window: ≤ 10 years from deposit.

8 Anticipated objections and responses

8.1 “This is too general to be empirical.”

The principle makes specific, falsifiable predictions enumerated in Sections 6 and 7. Generality is not the same as vacuity: Newton’s laws are extremely general but make specific predictions. RCE is general in the same sense; its operational content lies in the cross-domain predictions in Section 7 and the falsification conditions in Section 6.

8.2 “You are gluing together fields you don’t have credentials in.”

The author has no credentials in medicine, economics, or metrology. This is a real epistemic limitation and is acknowledged. The predictions in Section 7 are structural inferences from the principle, not domain-expert claims about each field’s substantive content. The appropriate response from domain experts is to evaluate whether the proposed regime architecture matches

the field’s actual coupling structure, and if not, to provide the corrected architecture. The principle stands or falls with the corrected architectures, not with the author’s draft versions.

8.3 “This is Ashby / Beer / Bateson / Friston restated.”

It is convergent with all of those, as Section 2.5 documents. What is new is the synthesis: the joint imposition of (a) cybernetic variety constraints, (b) frozen-parameter governance, (c) sealed pre-registered predictions, (d) a falsification ledger, and (e) explicit cross-domain isomorphism. None of the cybernetic precursors had (c)–(e). RCE provides an operational machine where they had a philosophical position.

8.4 “This is Bunge’s systemism / Laszlo’s systems philosophy redux.”

The objection is anticipated and serious. Several twentieth-century systemic-philosophy programmes attempted comparable syntheses — Bunge’s systemist ontology, Laszlo’s general systems philosophy, Cilliers’ work on complexity and postmodernism — and produced substantial philosophical literature without producing falsifiable empirical programmes. The pattern is consistent enough that any present-day attempt at cross-domain epistemic synthesis must explain why it does not converge to the same outcome.

The answer is structural, not rhetorical. The earlier programmes lacked four specific operational components that RCE inherits from EFC: (i) sealed numerical predictions with deposit hashes and pre-registered falsifiers; (ii) a living falsification ledger in which failed predictions are preserved with their failure modes documented; (iii) frozen-parameter governance enforced across regime boundaries; and (iv) external arbitration channels (Stage-IV surveys in cosmology, RCTs in medicine, out-of-sample crisis episodes in economics) whose outcomes cannot be narratively reabsorbed. Bunge, Laszlo, and Cilliers had philosophical positions; RCE has an operational machine. The distinction is empirical: a philosophy of science cannot be falsified by data; an operational governance framework instantiated in a specific predictive programme can be.

The risk is that RCE could collapse back to philosophical systemism if it is detached from the sealed-prediction infrastructure. Section 6’s falsification conditions for the principle itself, and Section 7’s pre-registered cross-domain predictions, are the operational anchors that prevent this collapse. If those anchors fail — if no sealed prediction in any domain ever resolves either way over the next decade — then the distinction from earlier systemic philosophies has not been operationally demonstrated and the present synthesis defaults to the same epistemic status as its predecessors.

8.5 “Universal pretensions in any field require extraordinary evidence.”

Acknowledged. The author makes no claim of universality. The principle predicts *homologous validation pathologies* in domains with coupled multi-regime structure, which is a more restricted claim than universal applicability. Domains without genuine scale separation, or with measurement architectures that already enforce regime-locked inference (e.g. controlled laboratory experiments in well-isolated single-regime systems), are explicitly outside the principle’s scope. The cross-domain predictions in Section 7 are confined to domains where the multi-regime structure is independently established.

9 Conclusion

The Energy-Flow Cosmology programme produced, over the period 2024–2026, a complete validation architecture: EBE, RCMP, the L0–L3 regime structure, frozen-parameter governance, the global parameter registry, sealed numerical predictions, the falsification ledger, the Framework Atlas with rival-mechanism mapping, and the operational ξ/η regime coordinates. The cosmological content of the programme — the entropy-flow account of gravitational modification, the survival valley $\mu \in [0.93, 0.96]$, $\Sigma \in [1.03, 1.07]$, the lensing crossover at $z \approx 0.44$ — is one application of that architecture.

This paper formalises the architecture itself as a principle of cross-domain epistemic structure: Regime-Closure Epistemology. The principle states that complex systems with coupled multi-regime dynamics admit valid validation only under the joint constraints of regime locality, variety closure, modelling closure, frozen-parameter governance, and external arbitration. The principle predicts that the “mystery entities” postulated across empirical domains (dark matter, idiopathic disease, exogenous shocks, systematic error of unknown origin, emergent capability) are not features of the systems being studied but artefacts of the measurement architectures deployed to study them.

The principle is convergent with prior independent work in the cybernetic tradition (Ashby 1956, Conant–Ashby 1970, Bateson 1972, Beer 1972), in second-order cybernetics (von Foerster 1974), and in the free-energy / active-inference tradition (Friston 2010). What this paper adds is the operational machinery for testing the principle in domains other than cosmology, via the sealed pre-registered predictions in Section 7 and the falsification conditions in Section 6.

The author makes no claim that RCE is correct. The claim is that the principle is sufficiently precise to be tested, the tests are declared, the failure conditions are explicit, and the falsification ledger is open. The same epistemic standard EFC has imposed on itself since 2024 is now imposed on its own generalisation.

If the principle holds, then a substantial fraction of contemporary empirical science is operating under a remediable structural pathology, and the dissolution of mystery entities across multiple domains is a tractable research programme rather than a generational mystery. If the principle fails, the failure will be documented in the ledger and the corpus will be revised accordingly. Either outcome advances the structure of validation in coupled multi-regime systems.

References and corpus links

1. Ashby, W. R. (1956). *An Introduction to Cybernetics*. Chapman & Hall, London.
2. Bateson, G. (1972). *Steps to an Ecology of Mind*. Chandler.
3. Beer, S. (1972). *Brain of the Firm*. The Penguin Press.
4. Bunge, M. (1979). *Treatise on Basic Philosophy, Volume 4: A World of Systems*. Reidel.
5. Cilliers, P. (1998). *Complexity and Postmodernism: Understanding Complex Systems*. Routledge.
6. Conant, R. C. & Ashby, W. R. (1970). Every good regulator of a system must be a model of that system. *International Journal of Systems Science*, 1(2), 89–97.
7. Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11, 127–138.
8. Goodhart, C. A. E. (1975). Problems of monetary management: the U.K. experience. Papers in Monetary Economics, Reserve Bank of Australia.
9. Laszlo, E. (1972). *Introduction to Systems Philosophy: Toward a New Paradigm of Contemporary Thought*. Gordon & Breach.
10. Magnusson, M. (2025). Energy-Flow Cosmology: A Thermodynamic Bridge Between General Relativity and Quantum Field. DOI [10.6084/m9.figshare.30402427](https://doi.org/10.6084/m9.figshare.30402427).
11. Magnusson, M. (2026). The Regime-Consistent Measurement Principle (RCMP): A Methodological Framework for Multi-Scale Physics. DOI [10.6084/m9.figshare.31222900](https://doi.org/10.6084/m9.figshare.31222900).
12. Magnusson, M. (2026). EBE Core Principles v1.0. DOI [10.6084/m9.figshare.31222903](https://doi.org/10.6084/m9.figshare.31222903).

13. Magnusson, M. (2026). L0–L3 Regime Architecture in Entropy-Bounded Empiricism. DOI [10.6084/m9.figshare.31112536](https://doi.org/10.6084/m9.figshare.31112536).
14. Magnusson, M. (2026). Validity-Aware AI: An Entropy-Bounded Architecture for Regime-Sensitive Inference. DOI [10.6084/m9.figshare.31122970](https://doi.org/10.6084/m9.figshare.31122970).
15. Magnusson, M. (2026). Foundations of Energy-Flow Cosmology (EFC): Regime Architecture and Methodological Principles of Entropy-Bounded Empiricism. DOI [10.6084/m9.figshare.31135597](https://doi.org/10.6084/m9.figshare.31135597).
16. Magnusson, M. (2026). Regime-Bound Measurement in Complex Systems: Proxy, Placement, and Validity. DOI [10.6084/m9.figshare.31564123](https://doi.org/10.6084/m9.figshare.31564123).
17. Magnusson, M. (2026). Regime-Based World Modeling: A Layered Epistemic Architecture for Complex Systems. DOI [10.6084/m9.figshare.31223650](https://doi.org/10.6084/m9.figshare.31223650).
18. Magnusson, M. (2026). Regime-Locked Measurement as a General Structural Constraint. DOI [10.6084/m9.figshare.31833076](https://doi.org/10.6084/m9.figshare.31833076).
19. Magnusson, M. (2026). KT3b Cross-Regime Measurement Failure. DOI [10.6084/m9.figshare.31963821](https://doi.org/10.6084/m9.figshare.31963821).
20. Magnusson, M. (2026). Cross-Domain Bridge Equations for the EFC Framework. DOI [10.6084/m9.figshare.31940547](https://doi.org/10.6084/m9.figshare.31940547).
21. Magnusson, M. (2026). Composite-Level Aggregation in Rotation Curve Model Selection. DOI [10.6084/m9.figshare.32162706](https://doi.org/10.6084/m9.figshare.32162706).
22. Magnusson, M. (2026). Methodological Convergence in Stage-IV Cosmology: Emergence of Epistemic Holism Under Precision Constraints. DOI [10.6084/m9.figshare.32080080](https://doi.org/10.6084/m9.figshare.32080080).
23. Magnusson, M. (2026). Ledger Epistemology and Regime-Closure Governance: An Operational Framework for Validation in Coupled Multi-Scale Systems (Version B, companion paper). DOI [10.6084/m9.figshare.32256951](https://doi.org/10.6084/m9.figshare.32256951).
24. Maturana, H. & Varela, F. (1980). *Autopoiesis and Cognition*. Reidel.
25. McEwen, B. S. (1998). Protective and damaging effects of stress mediators. *New England Journal of Medicine*, 338, 171–179.
26. Minsky, H. P. (1986). *Stabilizing an Unstable Economy*. Yale University Press.
27. Pearl, J. (2000). *Causality*. Cambridge University Press.
28. von Foerster, H. (1974). *Cybernetics of Cybernetics*. Biological Computer Laboratory, University of Illinois.

Version 1.0. This document is a deposit-ready preprint. Sealing protocol: the predictions in Section 7 are pre-registered against the DOI assigned at deposit. Falsification entries will be added to the EFC Validation Ledger and reflected in subsequent versions. Cross-links: RCE binds to RCMP (DOI 31222900) as the cosmological instantiation, to EBE (DOI 31222903) as the methodological substrate, to the L0–L3 architecture (DOI 31112536) as the typological foundation, and to Regime-Locked Measurement (DOI 31833076) as the cross-domain generalisation pilot. RCE supersedes the cross-domain framing of Regime-Locked Measurement while preserving its operational content.